QueryGenie

A Self-Correcting, Confidence-Aware Natural Language to SQL Interface

Ask a database in plain English — and get told when the answer can't be trusted

STUDENTS

- B.Tech CSIT
- Team 18

INNOVATION

- Confidence-aware abstention
- Self-correction

RESEARCH

- CodeS · SIGMOD 2024
- Spider benchmark

IMPACT

- Desktop app
- Runs locally

Abstract

Text-to-SQL models let people who do not know SQL ask a database questions in plain English, but they answer every question with the same confidence, so a wrong answer looks exactly like a right one. **QueryGenie** reproduces CodeS (SIGMOD 2024, arXiv:2402.16347), an open-source text-to-SQL model, and makes it say when it is unsure. We reproduce CodeS-1B to 77.8% execution accuracy on Spider (paper: 77.9%) and 69.5% on Spider-Realistic (paper: 70.9%). Our enhancement, confidence-aware abstention, runs the candidate queries the model already generates and checks whether they agree: when they don't, it asks the user instead of guessing. It answers 90.4% of questions and is right on 83.3% of them, with a third fewer wrong answers — a statistically significant gain. Everything runs locally in a desktop app, on open weights.

Objectives

1. Reproduce CodeS-1B on Spider within the ±5–10% tolerance, logging every run.
2. Ablate the CodeS pipeline to find which components matter.
3. Make the system say when it is unsure: confidence-aware abstention, tested statistically.
4. Measure the benchmark-to-reality gap on realistically worded questions.
5. Deliver a working desktop application that runs fully locally.

Key Innovations & Contributions

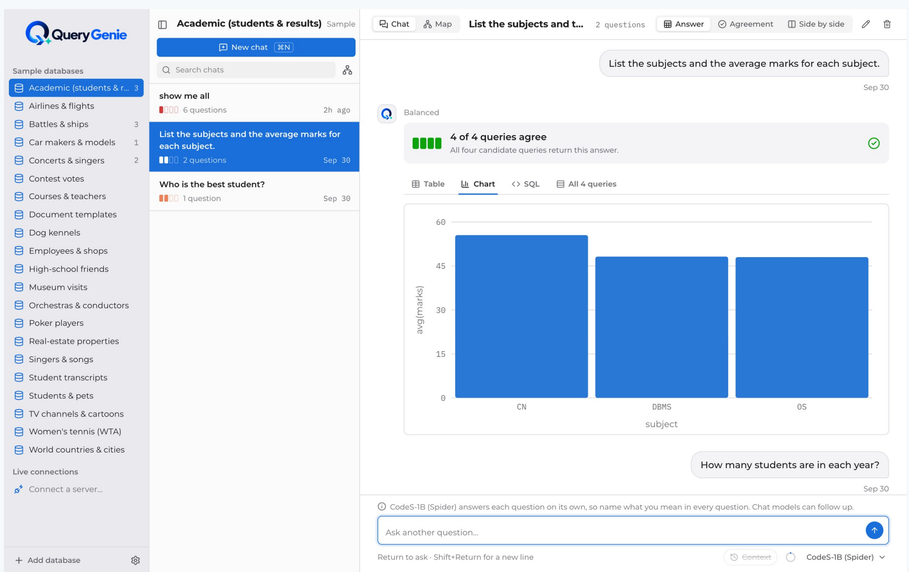
Confidence-aware abstention
Runs every candidate query, counts how many agree, and asks “Did you mean…?” when they don't.

Faithful, audited reproduction
77.8% vs 77.9% with the official evaluator, after an audit that fixed our own scoring and a value-retriever bug.

Execution-guided self-correction
Failed queries go back to the model with the database error; measured honestly (77.76 → 77.85%).

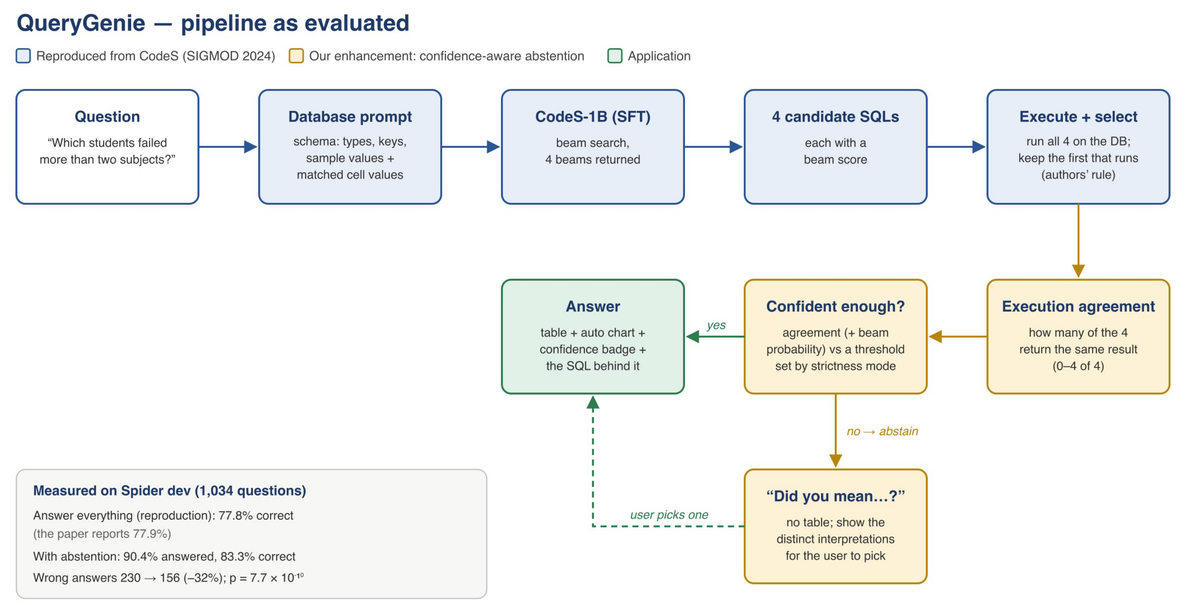
Local desktop application
Chats, schema map, charts, SQLite / PostgreSQL / MySQL — schema and data never leave the computer.

The Application



A confident answer: all four candidate queries agree, so QueryGenie shows the table and a chart.

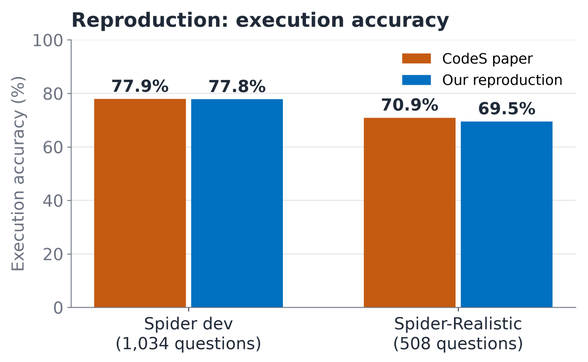
Proposed Methodology / System Architecture

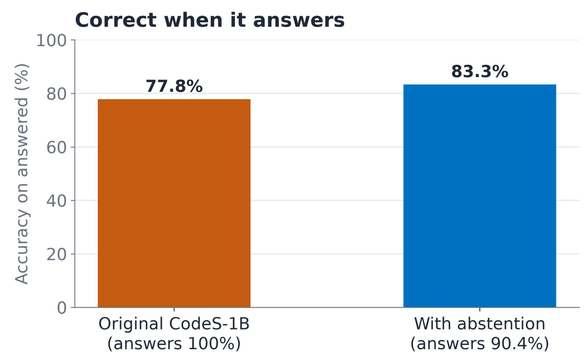


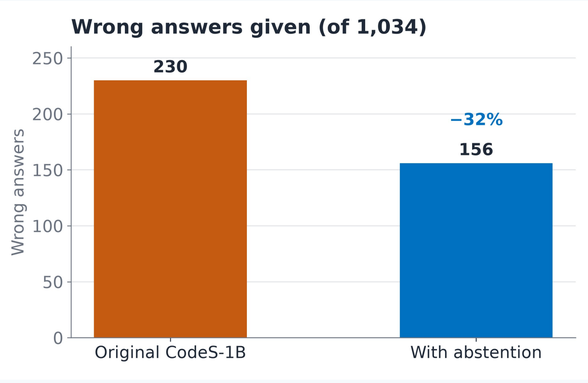
Datasets — all from Hugging Face (approved source; no Kaggle)

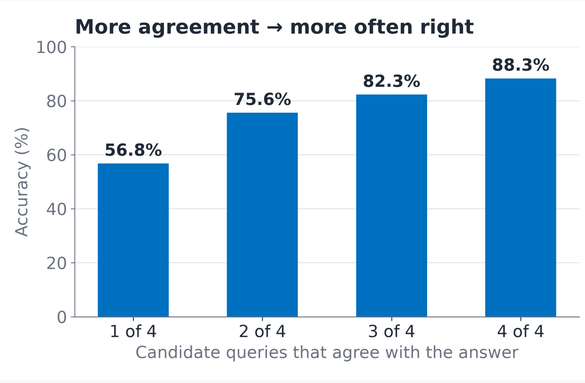
Set	Questions	Used for
Spider dev	1,034 · 20 databases	Reproduction, ablation, enhancement
Spider-Realistic	508	Realistic wording (no column names)
BIRD Mini-Dev	500 · 11 databases	Real-world databases (set up; future work)

Experimental Results









Key results

Reproduced: 77.8% EX on Spider (paper 77.9%), 69.5% on Spider-Realistic (paper 70.9%) — within tolerance.

Enhanced: answers 90.4% of questions, correct on 83.3% (vs 77.8%); wrong answers 230 → 156.

Significant: paired t-test $p = 7.7\times10^{-10}$, Wilcoxon $p = 9.8\times10^{-4}$, bootstrap 95% CI +4.5 to +6.7 pts.

Robust: counting each different query once, it still reaches 83.5% at 90% coverage.

Conclusion & Future Work

CodeS reproduced within 0.1 points; abstention makes its answers more trustworthy, running locally.

Next: CodeS-3B and more, more diverse candidate queries; then a platform that plugs into any database (Spark, Snowflake, NoSQL via a SQL gateway) with access control.

References

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[2] Yu, T., et al. (2018). Spider: A large-scale human-labeled dataset for complex and cross-domain semantic parsing and text-to-SQL task. arXiv:1809.08887.

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Team Details

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